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PatchCL-AE: Anomaly detection for medical images using patch-wise contrastive learning-based auto-encoder

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ABSTRACT

Anomaly detection is an important yet challenging task in medical image analysis. Most anomaly detection methods are based on reconstruction, but the performance of reconstruction-based methods is limited due to over-reliance on pixel-level losses. To address the limitation, we propose a patch-wise contrastive learning-based auto-encoder for medical anomaly detection. The key contribution is the patch-wise contrastive learning loss that provides supervision on local semantics to enforce semantic consistency between corresponding input–output patches. Contrastive learning pulls corresponding patch pairs closer while pushing non-corresponding ones apart between input and output, enabling the model to learn local normal features better and improve discriminability on anomalous regions. Additionally, we design an anomaly score based on local semantic discrepancies to pinpoint abnormalities by comparing feature difference rather than pixel variations. Extensive experiments on three public datasets (*i.e.*, brain MRI, retinal OCT, and chest X-ray) achieve state-of-the-art performance, with our method achieving over 99% AUC on retinal and brain images. Both the contrastive patch-wise supervision and patch-discrepancy score provide targeted advancements to overcome the weaknesses in existing approaches.

1. Introduction

Anomaly detection plays an important role in clinical disease screening. Disease screening helps to detect diseases early and timely treatment enables patients to control disease progression. Disease screening requires clinicians to review large volumes of medical images across modalities, yet the limited number of clinicians cannot meet the demand for large-scale screening. In addition, the screened population is mostly normal with a small portion of patients who may have diverse diseases. Although supervised deep learning methods (Zhang et al., 2023; Dolz et al., 2020) have achieved satisfactory performance in disease screening, these supervised methods require large amounts of annotated data, which is very time-consuming for clinicians to label (Zhao et al., 2021; Guo et al., 2023). In clinical practice, Magnetic Resonance Imaging (MRI), Computed Tomography (CT), and Optical Coherence Tomography (OCT) images contain dense multi-slice scans. It is extremely laborious for clinicians to annotate lesions in these multi-slice medical images. Moreover, manual annotations usually only cover some of the diseases. Many anomalies lack clinical annotation data needed for supervised learning. Therefore, supervised deep learning methods are limited by the quantity of annotations and can only detect classes of diseases contained in the training set. Compared with supervised methods, medical anomaly detection identifies deviations from normal medical data through unsupervised learning methods. These deviations indicate underlying health issues in patients. Medical Anomaly detection has been explored in the application of medical imaging (Guo et al., 2023; Zhou et al., 2021), including brain, fundus, and chest. Take brain MRI as an example, medical anomaly detection methods can detect abnormal regions in brain MRI slices without requiring clinicians to annotate various lesions in brain MRI slices. Therefore, anomaly detection is a promising technique to overcome the obstacles of limited labeled data and high annotation cost for medical image analysis (Baur et al., 2021; Zhou et al., 2021).

Existing anomaly detection methods can mainly be divided into two categories (Pang et al., 2021): image reconstruction-based approaches and non-image reconstruction based approaches. Reconstruction-based approaches have attracted many researchers due to their simplicity and feasibility. Image reconstruction-based approaches often utilize an auto-encoder to reconstruct the input data. During training, the model tries to minimize the reconstruction error, which is the difference between the raw input and the output of the model. Once training is completed, the model is used to reconstruct input data and calculate

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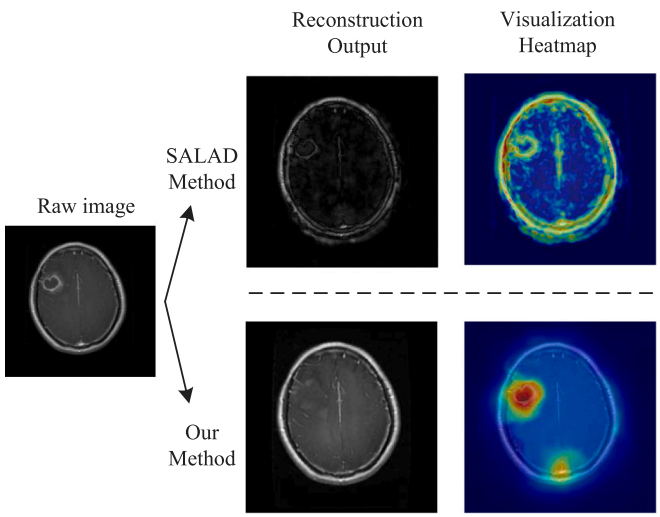

Fig. 1. Comparison of visualization results. The first row shows the reconstruction outputs and visualized heatmaps from the SALAD method. The second row presents the reconstruction outputs and heatmaps from the proposed approach.

the reconstruction error, where data with larger reconstruction errors are considered as anomalies (Chen et al., 2018).

This paper focuses on improving reconstruction-based anomaly detection approaches. Existing image reconstruction-based approaches have two main challenges: (1) They lack an understanding of normal local features, resulting in failure to reconstruct reasonable local features. (2) Pixel-level discrepancy-based anomaly scores are easily affected by pixel shifts, leading to erroneous anomaly detection. As shown in Fig. 1, we present reconstruction outputs and visualization heatmaps from the state-of-the-art (SOTA) method, *i.e.*, SALAD (Zhao et al., 2021). It can be observed that lesion information is still retained in SALAD's reconstructed outputs, which hinders the effective identification of the lesions. Moreover, the visualization heatmaps contain a lot of noise due to pixel shifts in the reconstructed outputs, which can be confusing for clinicians when using heatmaps to assist diagnosis.

To improve the ability to reconstruct reasonable local features, we propose a patch-wise contrastive learning-based auto-encoder. This approach aims to enhance the auto-encoder's intrinsic understanding of local features in normal samples, enabling improved reconstruction capabilities. Specifically, a patch-wise semantic consistency constraint between the input and output images is introduced into the auto-encoder. The patch-wise semantic constraint uses contrastive learning to make the image patches in the reconstructed image close to the corresponding input image patches in the deep feature space, while far away from the non-corresponding image patches in the input image. This constraint ensures that the reconstructed local semantic information is consistent with the style of normal samples.

Furthermore, to alleviate the issue of inaccurate anomaly detection caused by pixel-level noise, we propose a novel anomaly score based on local semantic feature difference rather than pixel-level error. Specifically, the reconstructed image (output) and the raw image (input) are fed into the encoder of the auto-encoder to generate compressed multi-level features, where the compressed features represent the deep semantics of the image. The compressed features of the reconstructed output are compared with those of the raw input in order to capture semantic discrepancies between them. Regions with large semantic deviations are identified as anomalous regions.

Our main contributions are summarized as follows:

1. To our best knowledge, this is the first work to employ a patch-level semantic consistency constraint for enhancing the local reconstruction capability of auto-encoders in medical anomaly detection. The consistency constraint enables the auto-encoder to reconstruct abnormal samples into reasonable normal ones by introducing the proposed patch-wise contrastive learning loss.
2. We propose a novel anomaly score based on local semantic discrepancy, which effectively addresses the negative impact of pixel-level noise on detection accuracy. By comparing the deeper semantic difference between input–output patches, this approach mitigates the influence of pixel-level noise and improves the overall detection performance.
3. The proposed approach achieves state-of-the-art anomaly detection results on retinal OCT, chest X-ray, and brain MRI datasets. This highlights the benefits of patch-wise contrastive learning.

2. Related work

In this section, we review works related to anomaly detection methods, introduce contrastive learning approaches, and compare our method to SOTA methods in terms of similarities and differences.

2.1. Anomaly detection

2.1.1. Image reconstruction-based approach

The core idea behind image reconstruction-based methods is to compress the image and then reconstruct it, using the difference between the original image and the reconstructed image as an indicator to determine anomalies. Lu and Xu (2018) used reconstruction probabilities from a variational auto-encoder for medical anomaly detection. Schlegl et al. (2017) tried to introduce a generative adversarial network (GAN), AnoGAN, for anomaly detection, which utilizes a GAN to learn normal anatomical variability and then computes the distance between synthesized images from the learned distribution and test images to estimate the likelihood of anomalies. Further, an improved version of AnoGAN, f-AnoGAN (Schlegl et al., 2019), was proposed, which introduces an additional encoder to map test images to a latent space instead of relying on feature matching in the latent space in the method of AnoGAN. Ganomaly (Akcay et al., 2019) uses an adversarial training framework for semi-supervised anomaly detection. It trains an encoder and discriminator on normal samples only, reconstructing normal and discriminating anomalous inputs. Zhao et al. (2021) proposed latent space consistency constraints and central loss to improve the compactness of the latent space for improving the performance of GAN-based methods.

Unlike existing reconstruction-based approaches, our method incorporates a local consistency constraint during training to align the distributions of local features between the input and output. This constraint enables the auto-encoder to learn the distribution of normal local patch features better. Additionally, the proposed anomaly score calculation based on local semantic discrepancy differs from previous pixel-level semantic comparisons. This new approach allows for more robust anomaly detection, as it is less sensitive to pixel-level noise and variations.

2.1.2. Non-image reconstruction-based approach

Non-image reconstruction approaches mainly include one-class classification approaches and deep feature embedding approaches. By training on normal examples only, **one-class classification approaches** seek to create a boundary or decision function that effectively separates the normal class from the anomalous ones. One-class support vector machine (OCSVM) (Schölkopf et al., 2001) and deep SVDD (Ruff et al., 2018) are typical examples of one-class classification models. Deep SVDD is an extension of the SVM approach based on deep learning techniques. It utilizes a neural network to map input data into a high-dimensional feature space and learns a hypersphere in this space that encapsulates the normal class. The data outside the hypersphere are called anomalies. Further, a modified version of deep SVDD, patch SVDD (Yi and Yoon, 2020), is proposed, which operates on image

patches rather than full images. It partitions an image into overlapping patches and treats each patch as an individual data point for SVDD modeling. This approach captures local texture and structural information within an image, making it suitable for anomaly detection in image data. **Deep feature embedding-based approaches** are proposed to overcome the limitation of insufficient feature-level discriminatory information. The knowledge distillation framework is a good choice to achieve feature-level information discrimination, where the teacher and student models in the framework compare deep features embedding for anomaly detection. Bergmann et al. (2020) are the first to utilize a distillation framework for anomaly detection, where multiple student networks are trained to regress the outputs of a teacher network. When the outputs of the student networks differ from the output of the teacher network, the input samples are considered anomalous. Additionally, the uncertainty among multiple student networks is used as an additional score function to indicate anomalies. Further, MKD and STFM methods (Salehi et al., 2021; Wang et al., 2021) showed that multiple intermediate feature learning in distillation leads to better exploitation of the teacher's knowledge and a more distinctive discrepancy between teacher and student networks, compared with utilizing the last layer activation values only. Deng and Li (2022) propose RD4AD to further improve the performance of the distillation framework, in which a reverse distillation strategy is introduced.

2.2. Contrastive learning

Contrastive learning has emerged as a powerful paradigm for representation learning across domains. Early contrastive methods like LSTMC (Srivastava et al., 2015) define a discriminative loss to distinguish positive sample pairs from negative samples. Deep InfoMax (Hjelm et al., 2018) maximizes the mutual information between global and local features. More recent contrastive techniques leverage data augmentations to construct positive pairs and negative pairs for comparison. Notable examples include CPC (van den Oord et al., 2019) using sequential predictions, SimCLR (Chen et al., 2020a) with large batch sizes and data augmentations, and MoCo (He et al., 2020a) introducing a dynamic dictionary and momentum encoder.

Contrastive objectives have evolved from using random negatives to more meaningful anchor representations for comparison. Cross-entropy, triplet margin, and clustering losses have been explored for optimization. Self-supervised contrastive learning has become a prevalent approach for pre-training visual models. Contrastive methods have achieved state-of-the-art results on many representation learning benchmarks. However, contrastive learning has not been effectively tailored for medical anomaly detection tasks yet. Our work successfully exploits the representation learning capabilities of contrastive learning for this domain by incorporating patch-wise contrastive objectives into a reconstruction-based framework.

3. Method

Problem: Given a training set of normal samples $D_{train} = \{x_i, i = 1, \dots, n\}$ and a testing set $D_{test} = \{(x_j, y_j), j = 1, \dots, m\}$, where x_i or x_j represents input sample and $y_i \in \{0, 1\}$ represents label, with 0 denoting normal and 1 denoting anomaly. The model f is trained on the training set D_{train} containing only normal samples. During testing, the model will be evaluated on D_{test} to differentiate normal and anomaly samples. If the model f can reconstruct the normal input image with a small reconstruction error and abnormal ones with a large reconstruction error, it would achieve satisfactory anomaly detection performance.

To this end, we propose a patch-wise contrastive learning-based auto-encoder (patchCL-AE) to introduce local semantic supervision for increasing the local reconstruction error of abnormal input images. The proposed local-level supervision enables the auto-encoder to learn the normal local patch distribution better, making the auto-encoder reconstruct anomalous regions in abnormal samples into normal regions, thus resulting in larger reconstruction errors for abnormal samples. Additionally, to mitigate the impact of pixel-level shifts in the reconstructed output on pixel-level anomaly scores, we propose an anomaly score based on patch-wise semantic difference between the input and output.

3.1. Patch-wise contrastive learning-based auto-encoder (patchCL-AE)

As shown in Fig. 2, the proposed patchCL-AE consists of three parts during the training process: an auto-encoder, a global image loss, and a local patch loss. The encoder–decoder is used for image reconstruction. The global loss and local loss optimize the reconstruction from perspective of global image and local patch, which are presented in the green and red boxes in Fig. 2, respectively. Specifically, a raw image X is first added with gaussian noise and then fed into the auto-encoder (AE). The goal of AE is to generate a reconstructed image $\hat{X}$ similar to the raw image X. Similar to the previous work (Zhao et al., 2021), we take the adversarial loss as the global image loss to supervise the global image style in the reconstructed image. However, image-level losses are insufficient in reconstructing local semantic information, as mentioned in the introduction. To alleviate this issue, we employ patch-wise contrastive learning as a local loss to enforce the consistency of local feature distributions between the input and output, while enabling the model to learn the local manifold of normal samples. During the testing process, we calculate the difference between the input and reconstructed output using the proposed patch-wise anomaly score, as shown in Fig. 4.

3.1.1. Auto-encoder

Similar to the classic convolutional AE, the encoder and decoder are used to compress and reconstruct image information respectively in the proposed patchCL-AE. The encoder of the AE contains five convolutional blocks. Each block consists of multiple 3×3 convolutional kernels with a stride of 1 and padding of 1, followed by a pooling layer at the end. In addition, each convolutional layer is followed by an InstanceNorm layer and a ReLU activation layer by default. The decoder of the AE contains four convolutional blocks as well. In each convolutional block, there is a linear upsampling layer. The detailed structure of the encoder and decoder is shown in Table 1.

3.1.2. Image-wise loss

Following the previous method (Zhao et al., 2021), An adversarial loss is utilized as the global image loss for image reconstruction. The global image loss encourages the reconstructed image to be visually similar to the raw image. Image-wise loss $\mathcal{L}_{Img}$ is formulated as follows:

$$\mathcal{L}_{Img}(E, De, D) = \mathbb{E}_X \log D(X) + \mathbb{E}_X \log(1 - D(\hat{X})), \tag{1}$$

where $\hat{X} = De(E(X))$ represents the reconstructed output of X, E represents the encoder, De represents the decoder, D represents the discriminator, and $\mathbb{E}$ represents the mathematical expectation. The detailed structure of the discriminator is shown in Table 2.

3.1.3. Patch-wise loss via contrastive learning

Patches reflect local semantic and texture information. Our goal is to make the patches in the reconstructed image obtain the same information as the corresponding patches in the raw image. Although pixel-wise loss partially achieves this, it fails to comprehend the semantic information at the local region. The pixel-wise loss solely focuses on minimizing the mean square error without capturing the underlying semantic content. To overcome this issue, patches are fed into the encoder of patchCL-AE to obtain deep semantic features, as shown in Fig. 2. Feature representations of corresponding patches between the input and output are pulled closer together via contrastive learning,

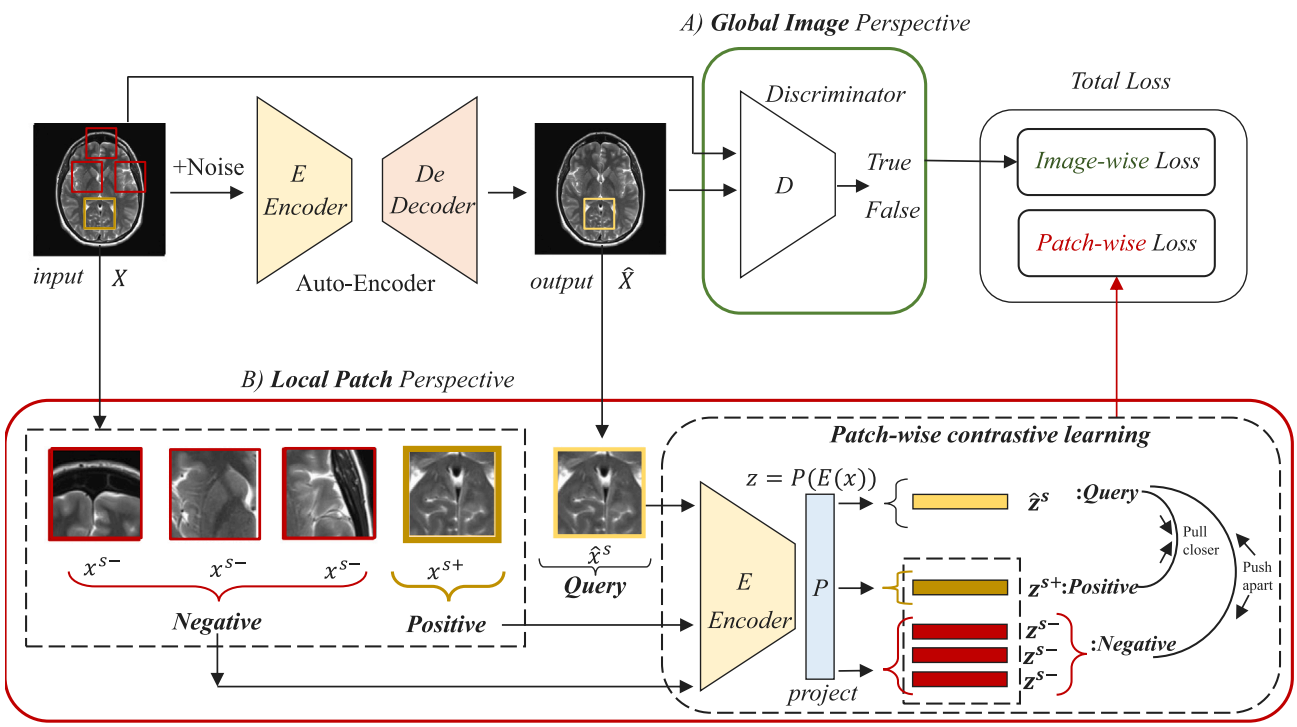

Fig. 2. Overview of the proposed patchCL-AE approach. The patchCL-AE comprises an Encoder-Decoder and two loss functions from global and local perspectives, *i.e.*, global image-wise loss and local patch-wise loss. The global image-wise loss employs an adversarial learning objective, while the local patch-wise loss utilizes contrastive learning between patch image pairs.

while the features of non-corresponding patches are pushed further apart.

The encoder E in the auto-encoder is utilized to encode the raw image X and the reconstructed image $\hat{X}$, where the patch images are compressed to hierarchical features. The information contained in features from different layers and spatial locations can be further exploited through contrastive learning. Although contrastive learning does not require annotated data, we need to define positive samples and negative samples for contrastive training. Optimization of contrastive learning is achieved by increasing the distance between the query and negative samples while reducing the distance between the query and positive samples. However, the selection of positive and negative samples in popular contrastive learning methods (Chen et al., 2020b; He et al., 2020b) is not suitable for our approach. Therefore, we redefine the concepts of query, positive, and negative samples in our approach.

Query, positive, and negative samples. As shown in Fig. 2, we define the patches in the reconstructed output as **query** samples and the corresponding patches in the raw input as **positive** samples. Different from positive samples, the patches in the raw input that do not correspond to the query patches are considered as **negative** samples. These patches x are encoded and further mapped by the encoder E and projection head P (two fully-connected layers) to features z in a high-dimensional space. The detailed structure of the projection is shown in Table 3. Specifically, we calculate these features z as shown in Fig. 3. **Query** $\hat{\boldsymbol{z}}^s$: the patch $\hat{\boldsymbol{x}}^s$ from the output image $\hat{X}$ will generate a set of features $\{\hat{\boldsymbol{z}}_l^s\}_L$, where $\hat{z}_l^s$ represents the features from spatial location s in the lth layer generated by encoder E and project P, and L denotes the number of layers. **Positive** $\boldsymbol{z}^{s+}$: Corresponding features from the input patch image $\boldsymbol{x}^{s+}$ are denoted as $\{\boldsymbol{z}_l^{s+}\}_L$. **Negative** $\boldsymbol{z}^{s-}$: Other Non-corresponding features from the input patch image $\boldsymbol{x}^{s-}$ are denoted as $\{\boldsymbol{z}_l^{s-}\}_L$. We formulate $\{\hat{\boldsymbol{z}}_l^s\}_L$, $\{\boldsymbol{z}_l^{s+}\}_L$, and $\{\boldsymbol{z}_l^{s-}\}_L$ as follows:

$$\begin{cases} \{\hat{\boldsymbol{z}}_l^s\}_L &= \{P_l\left(E_l\left(\hat{x}^s\right)\right)\}_L \\ \{\boldsymbol{z}_l^{s+}\}_L &= \{P_l\left(E_l\left(\boldsymbol{x}^{s+}\right)\right)\}_L \\ \{\boldsymbol{z}_l^{s-}\}_L &= \{P_l\left(E_l\left(\boldsymbol{x}^{s-}\right)\right)\}_L \end{cases}, \tag{2}$$

where $E_l(*)$ represents the output of the lth layer in the encoder, and P_l represents the corresponding projection layer.

Compute the distance between the query and the positive/negative samples. Following the contrastive learning method (He et al., 2020b), we scale the cosine similarity between features by a temperature τ and further pass it as logit. A cross-entropy loss is utilized to optimize the similarity between corresponding features from the raw input patch and the reconstructed output patch. Then the **patch-wise loss** is defined as

$$\begin{aligned} \mathcal{L}_{Patch} &= \sum_{l=1}^{L}\sum_{s=1}^{S_l} \ell\left(\hat{z}_l^s, \boldsymbol{z}_l^{s+}, z_l^{s-}\right) \\ &= \sum_{l=1}^{L}\sum_{s=1}^{S_l} -\log\left[\frac{e^{sim(\hat{z}_l^s, z_l^{s+})/\tau}}{e^{sim(\hat{z}_l^s, z_l^{s+})/\tau} + \sum_{z_l^{s-}} e^{sim(\hat{z}_l^s, z_l^{s-})/\tau}}\right], \end{aligned} \tag{3}$$

where S_l denotes the set of all spatial locations s in the lth layer.

3.1.4. Total loss

Our final loss incorporates image-wise and patch-wise losses as follows.

$$\mathcal{L}_{Total} = \mathcal{L}_{\text{Img}}(E, De, D) + \mathcal{L}_{\text{Patch}}(E, De), \tag{4}$$

The image-wise loss is used to constrain the style consistency between the input and output images. The patch-wise loss is used to constrain patch regions in the output, making the output image match the raw input image more closely in local details. The detailed optimization method of the proposed patchCL-AE is described in Algorithm 1.

3.2. Testing process and anomaly score

During the testing process, the raw input X is first fed into the auto-encoder to generate the reconstructed output $\hat{X}$. Then, the difference between the raw input X and the reconstructed output $\hat{X}$, *i.e.*, the anomaly score, is calculated. Unlike previous reconstruction-based

Table 1
The network structure of the Encoder (E) and Decoder (De).

	Layer name	Output size	Operations	Param#
E	E_1	224 × 224	conv 7 × 7,64 InstanceNorm ReLU	9,921
	E_2	112 × 112	conv 3 × 3,128 InstanceNorm ReLU Downsample	74,305
	E_3	56 × 56	conv 3 × 3,256 InstanceNorm ReLU Downsample	295,393
	E_4	28 × 28	conv 3 × 3,256 InstanceNorm ReLU Downsample	590,193
	E_5	28 × 28	[conv 3 × 3, 256 InstanceNorm ReLU conv 3 × 3, 256 InstanceNorm ReLU] × 6	7,081,644
De	De_4	56 × 56	upsample conv 3 × 3,256 InstanceNorm ReLU	590,137
	De_3	112 × 112	upsample conv 3 × 3,128 InstanceNorm ReLU	295,153
	De_2	224 × 224	upsample conv 3 × 3,64 InstanceNorm ReLU	74,017
	De_1	224 × 224	conv 7 × 7,3 Tanh	9,411

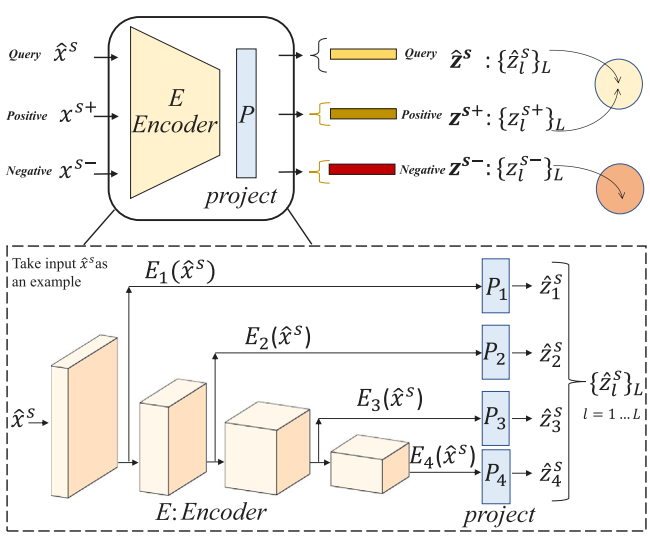

Fig. 3. The calculation process of deep features.

methods that use pixel-level difference as an anomaly score, we propose a patch-wise anomaly score based on the local semantic discrepancy between the input and output.

3.2.1. Patch-wise anomaly score

As shown in Fig. 4, the raw input X and reconstructed output $\hat{X}$ are fed into the encoder E and projection layer P to obtain deep semantic features f and $\hat{f}$, respectively. The semantic features of $\hat{X}$ and X are formulated as follows:

$$\begin{cases} \hat{\boldsymbol{f}}_l = P_l(E_l(\hat{X})), \\ \boldsymbol{f}_l = P_l(E_l(X)), \end{cases} \tag{5}$$

where $\hat{f}_l$ denotes the features of $\hat{X}$ after passing through the first layer to the lth layer of the encoder (E_l) and the corresponding projection head (P_l). Similarly, $\boldsymbol{f}_l$ represents the features generated after feeding the raw image X into the encoder and projection head.

We utilize cosine similarity to calculate the distance between features f and $\hat{f}$ as an anomaly score. The lth anomaly score map is calculated as follows:

$$\mathcal{M}_l = 1 - sim\left(\hat{\boldsymbol{f}}_l(h,w), \boldsymbol{f}_l(h,w)\right), \tag{6}$$

where (h, w) represents the spatial position of deep semantic features.

Since anomaly score maps from different layers have varying scales, all the anomaly score maps are scaled to the raw image size. The final anomaly score map is obtained by the summation of all the anomaly score map, as follows:

$$\mathcal{M} = \sum_{l=1}^{L} \left(resize(\mathcal{M}_l)\right) \tag{7}$$

3.2.2. Our anomaly score V. S. previous ones

Anomaly scores based on reconstruction methods can be classified into two categories: pixel-level reconstruction error and bottleneck feature-based difference. Pixel-level reconstruction error primarily focuses on the changes in individual pixels, while bottleneck feature difference captures the semantic changes in the entire image. Compared with these two approaches, our patch-wise anomaly score emphasizes the local semantic difference between the input and output

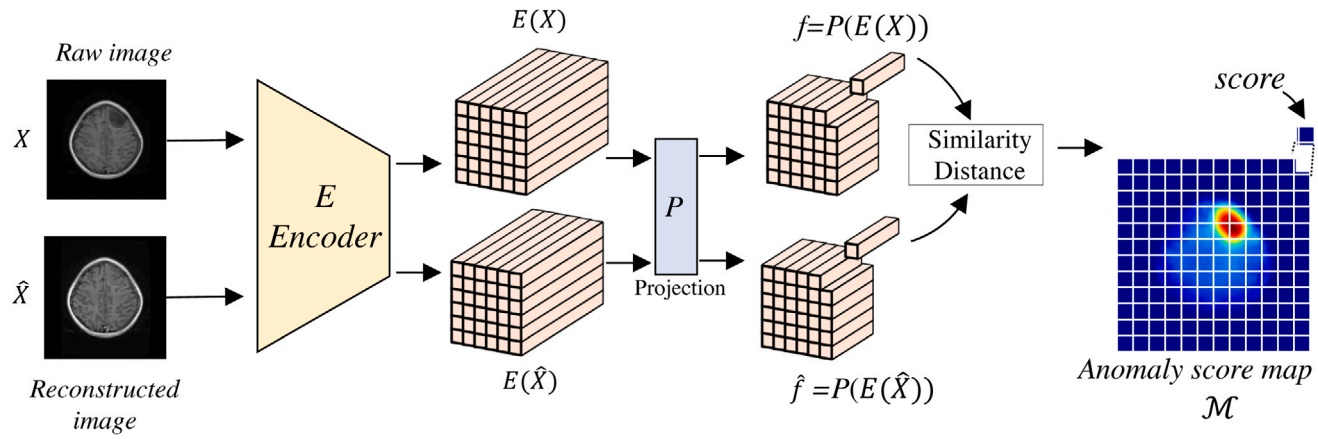

Fig. 4. The calculation process of anomaly score map. Both the raw image and the reconstructed image are fed into the encoder E to obtain features $E(X)$ and $E(\hat{X})$. The features are further mapped into deep semantic features f and $\hat{f}$ through the projection layer P. The anomaly score map is generated by calculating the similarity distance between features f and $\hat{f}$.

Table 2
The network structure of the Discriminator (D).

Layer name	Output size	Operations	Param#
D_1	112 × 112	conv $3 \times 3, 64$ InstanceNorm LeakyReLU Downsample	2,241
D_2	56 × 56	conv $3 \times 3, 128$ InstanceNorm LeakyReLU Downsample	131,425
D_3	28 × 28	conv $3 \times 3, 256$ InstanceNorm LeakyReLU Downsample	524,657
D_4	14 × 14	conv $3 \times 3, 512$ InstanceNorm LeakyReLU Downsample	2,097,721
D_5	7 × 7	conv $3 \times 3, 512$ InstanceNorm LeakyReLU Downsample	4,194,845
D_6	4 × 4	conv $3 \times 3, 512$ InstanceNorm LeakyReLU Downsample	4,194,845
D_7	8192	flatten	0
D_8	256	256d-FC	2,097,408
D_9	1	1d-FC	257

Table 3
The network structure of the Projection (P).

Layer name	Output size	Operations	Param#
P_1	256	256d-FC ReLU	65,792
P_2	256	256d-FC	65,792

patches. Furthermore, the proposed patch-wise anomaly scores incorporate multi-layer comparisons, where deeper levels correspond to larger receptive fields. Thus, the patch-wise anomaly score combines difference in multi-scale semantic information between the input and output, enabling more precise anomaly detection.

4. Experiments

4.1. Datasets

Three publicly available datasets, i.e., retinal OCT, chest X-ray, Brain MRI datasets, are adopted to evaluate the proposed patchCL-AE approach.

4.1.1. Retinal OCT dataset

It is an optical coherence tomography (OCT) dataset (Kermany et al., 2018), where the data contains four categories: choroidal neovascularization (CNV), diabetic macular edema (DME), drusen and normal. The training and testing sets have been pre-defined by the publishers to ensure fair comparison. As an unsupervised algorithm, our method

Algorithm 1 The training of the proposed PatchCL-AE

1: **for** number of training iterations **do**
2: Sample minibatch of m examples $\{X^{(1)}, \ldots, X^{(i)}, \ldots, X^{(m)}\}$ from normal medical images
3: Encoder (E) and decoder (De) generate reconstructed images $\hat{X}^{(i)} = De(E(X^{(i)} + \text{Noise}))$
4: Extract image patches $(x^{s+}, x^{s-}, \hat{x}^{s})$ from original image $X^{(i)}$ and corresponding reconstructed image $\hat{X}^{(i)}$
5: The image patches $(x^{s+}, x^{s-}, \hat{x}^{s})$ are projected onto a 256-dimensional feature space to generate feature vectors $(z^{s+}, z^{s-}, \hat{z}^{s})$ through the encoder (E) and projection head (P).
6: Update the discriminator (D) by ascending its gradient:

$$\nabla_{\theta_D} \frac{1}{m} \sum_{i=1}^{m} \left[\log D\left(X^{(i)}\right) + \log\left(1 - D\left(De\left(E\left(X^{(i)} + \text{Noise}\right)\right)\right)\right)\right].$$

7: Update the encoder (E), decoder (De), and projection (P) by descending its gradient:

$$\nabla_{\theta_{E,De}} \frac{1}{m} \sum_{i=1}^{m} \log\left(1 - D\left(De\left(E\left(X^{(i)} + \text{Noise}\right)\right)\right)\right) + \nabla_{\theta_{E,De,P}} L_{\text{patch}}\left(x^{s+}, x^{s-}, \hat{x}^{s}\right).$$

8: **end for**

only utilizes the normal class from the training set during training, and in the testing process, we evaluate its performance on the entire testing set.

4.1.2. Chest X-ray dataset

A dataset of chest X-ray images (Kermany et al., 2018) was collected specifically for pediatric cases. In the training set, 3883 images were classified as depicting pneumonia, of which 2538 were caused by bacterial infection and 1345 were caused by viral infection. In addition, 1349 images were classified as normal. These images were collected from 5856 patients, ensuring a wide range of cases for training models. The test set consists of 234 normal images and 390 pneumonia images (242 bacterial and 148 viral). Chest X-ray images, compared with OCT images, are more complex as they encompass multiple anatomical structures such as the lungs and bones. These images pose a greater challenge for detecting abnormalities accurately since lesions can manifest in various locations within the chest X-ray images.

4.1.3. Brain MRI dataset

The brain tumor MRI dataset (Cheng, 2017), was utilized for brain tumor classification in a Kaggle challenge. This dataset comprises 7,022 human brain MRI images that are categorized into four classes: glioma, meningioma, pituitary, and no tumor. The dataset has been divided into training and test sets to facilitate a comprehensive evaluation of the algorithm's performance.

4.2. Experimental setup

4.2.1. Implementation details

Our method is implemented based on PyTorch Deep Learning framework. The Adam optimizer is used to optimize the model parameters, where the initial learning rate is set as 0.002, and other parameters are set as $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\varepsilon = 0.0001$. The scale of the original image is resized to 256×256. The maximal number of model epoch for training is 400. For each layer's features, we sample 256 random locations, and apply 2-layer fully connected layer to acquire 256-dim final features.

4.2.2. Evaluation metrics

For performance evaluation, we calculate the area under curve (AUC), F1-score, average classification accuracy (ACC), sensitivity (SEN), and specificity (SPE) as the evaluation metrics. SEN and SPE are used to evaluate the performance of the model on positive samples and negative samples respectively; AUC, ACC, and F1-score are used to comprehensively evaluate the performance of the model on positive and negative samples. Here, F1-score, ACC, SEN, and SPE are defined as $F1\text{-}score = \frac{2TP}{2TP+FN+FP}$, $ACC = \frac{TP+TN}{TP+TN+FP+FN}$, $SEN = \frac{TP}{TP+FN}$, and $SPE = \frac{TN}{TN+FP}$, where TP, TN, FN, and FP are the true positives, true negatives, false negatives, and false positives, respectively. The thresholds used to calculate the evaluation metrics are chosen based on the best F1-score.

4.3. Comparison with state of the art (SOTA)

In our experiments, apart from the SOTA reconstruction-based approaches, including deep SVDD (Ruff et al., 2018), AE (Baur et al., 2019), VAE (Lu and Xu, 2018), f-AnoGAN (Schlegl et al., 2019), patch SVDD (Yi and Yoon, 2020), GANomaly (Akcay et al., 2019), and SALAD (Zhao et al., 2021), we also compare our method with non-reconstruction-based approaches including STFPM (Wang et al., 2021), Multiresolution Knowledge Distillation (MKD) (Salehi et al., 2021), and RD4AD (Deng and Li, 2022).

4.3.1. Overall anomaly detection performance

We evaluate the performance of our method and state-of-the-art methods on retinal OCT, chest X-ray, and brain MRI datasets. Our method and the state-of-the-art methods are trained 5 times using different random seeds. We list the mean results of the compared methods in Table 4. The mean and standard deviation of our method are shown in the last column of Table 4. We further analyze the significant differences between our method and the compared methods in AUC using p-values. The p-values are all less than 0.05 as shown in Table 5. The experimental results demonstrate that our method significantly outperforms the compared methods. Overall, the methods incorporating feature-level constraints (*i.e.*, SALAD, MKD, STFPM, and RD4AD) outperformed those without these constraints across all datasets significantly, demonstrating powerful anomaly detection capabilities. In particular, recent approaches such as SALAD and MKD further widened the performance gap compared with methods only using pixel-level or image-level constraints (*i.e.*, AE, VAE, Ganomaly, and f-AnoGAN). This demonstrates that constraints based on pixel-level or image-level are insufficient.

With the aid of feature-level constraint from the latent feature space, the SALAD model yields a satisfactory anomaly detection performance. Unlike the SALAD method, we enforce consistency between corresponding input–output patches using contrastive learning, thereby achieving superior detection performance across the three datasets. Specifically, on the retinal OCT dataset, our method achieves the best AUC, F1-score, ACC, SEN, and SPE scores of 99.31%, 99.20%, 98.80%, 99.86%, and 95.60%, respectively. Similar trends can also be observed on brain MRI and chest X-ray datasets.

Compared with the results on the retinal OCT and brain MRI datasets, the anomaly detection performance of the listed methods consistently decreases on chest X-ray dataset. The performance degradation is due to the greater complexity of content in chest X-ray

Table 4
Performance (AUC(%), F1-score (%), ACC (%), SEN (%), and SPE (%)) yielded by different algorithms on Retinal OCT, Brain MRI, and Chest X-ray datasets.

Datasets	Metrics	DeepSVDD	AE	VAE	Ganomaly	Patch SVDD	f-AnoGAN	SALAD	STFPM	MKD	RD4AD	Ours
Brain MRI	AUC	74.45	78.57	91.67	84.36	91.12	92.53	94.39	93.42	93.31	87.42	**99.20** ± 0.49
	F1-score	83.24	85.89	93.69	91.45	91.37	92.34	95.78	93.32	92.94	89.65	**98.03** ± 0.26
	ACC	75.59	79.12	91.45	86.93	88.02	89.53	94.32	90.09	89.98	84.61	**97.24** ± 0.38
	SEN	87.74	92.86	97.02	92.35	91.83	93.40	97.87	93.98	95.12	95.62	**98.49** ± 0.40
	SPE	48.40	49.03	79.12	77.53	79.52	80.21	86.43	82.32	79.03	62.32	**94.41** ± 0.23
Retinal OCT	AUC	78.21	77.80	81.26	84.15	84.72	84.23	96.32	96.09	96.10	97.66	**99.29** ± 0.12
	F1-score	88.21	85.92	85.99	89.54	88.39	85.21	93.40	95.68	93.98	96.42	**99.16** ± 0.20
	ACC	80.65	78.46	77.82	81.79	81.64	77.78	90.56	92.78	91.45	94.75	**98.76** ± 0.40
	SEN	96.81	94.20	95.31	95.90	93.46	89.93	95.71	95.49	97.59	95.96	**99.87** ± 0.48
	SPE	31.99	42.23	38.65	40.16	50.12	49.69	78.03	85.97	73.40	90.06	**95.46** ± 1.20
Chest X-ray	AUC	67.10	60.27	62.43	67.34	65.13	75.76	82.78	84.91	85.95	78.65	**91.14** ± 0.62
	F1-score	77.34	77.46	77.65	80.40	78.47	81.34	84.25	82.47	85.54	80.71	**88.60** ± 0.54
	ACC	63.95	64.48	66.75	71.19	68.26	74.33	76.67	78.28	80.02	72.51	**86.77** ± 0.79
	SEN	98.56	**98.81**	98.16	94.78	92.56	88.79	88.49	81.31	93.48	89.92	92.78 ± 0.65
	SPE	64.10	6.97	8.98	31.12	27.80	40.23	57.87	73.89	56.22	43.12	**74.90** ± 1.80

Table 5
Statistical analysis of the classification performance AUC between our method and the compared methods is conducted using paired t-tests on brain MRI, retinal OCT, and chest X-ray datasets.

Datasets		DeepSVDD	AE	VAE	Ganomaly	Patch SVDD	f-AnoGAN	SALAD	STFPM	MKD	RD4AD
Brain MRI		4.5×10^{-7}	9.3×10^{-7}	2.5×10^{-5}	2.0×10^{-6}	2.3×10^{-5}	4.6×10^{-5}	7.1×10^{-5}	1.5×10^{-4}	1.0×10^{-4}	5.9×10^{-6}
Retinal OCT	Ours	8.7×10^{-7}	8.1×10^{-7}	1.6×10^{-6}	3.3×10^{-6}	3.9×10^{-6}	3.4×10^{-6}	2.3×10^{-3}	1.7×10^{-3}	1.8×10^{-3}	2.3×10^{-2}
Chest X-ray		2.7×10^{-7}	1.0×10^{-7}	1.4×10^{-7}	2.9×10^{-7}	2.0×10^{-7}	1.6×10^{-6}	1.8×10^{-5}	5.9×10^{-5}	1.2×10^{-4}	3.7×10^{-6}

Table 6
Other evaluation metrics.

Method	Parameter size	FLOPs	Train-time	Infer-time
MKD	1.5×10^{7}	160G	51.34 min	394.2 ms
Ours	8.9×10^{7}	35G	15.10 min	113.7 ms

images. Unlike retinal OCT images and brain MRIs where lesions have clear boundaries, the abnormal pneumonia areas in chest X-rays are very similar to normal areas without distinct borders. This makes it challenging for reconstruction-based and feature embedding-based methods to differentiate between anomalies and normals. However, the proposed patchCL-AE achieves the best performance on the chest X-ray dataset, obtaining the highest AUC, F1-score, ACC, and SPE. This demonstrates that the proposed patch-wise contrastive learning can help the model understand semantic information in images better and discover subtle difference between normal and abnormal regions. In addition, we also compared with the SOTA method on other metrics including the number of parameters, floating point operations (FLOPs), train-time, and infer-time in Table 6.

4.3.2. Heatmap visualization results

We visualize the interpretability heatmaps produced by SOTA methods and the proposed approach on these three datasets. As illustrated in Fig. 5, the basic AE method only detects some lesion boundaries, while also erroneously highlighting edges of normal tissues as anomalies. Despite achieving satisfactory quantitative results in Table 4, SALAD over-detects numerous anomalous points in the visualizations since it uses pixel-wise anomaly score. The multi-feature distillation approach MKD covers lesion areas in the heatmaps but lacks precise localization. Compared with SOTA methods, the proposed patchCL-AE demonstrates improved lesion localization capabilities, especially when compared with reconstruction-based approaches. By leveraging patch-wise comparisons, our method effectively suppresses false positives from pixel-wise difference. This indicates the proposed patch-wise comparisons can better compare local semantic discrepancies rather than just pixel-wise value difference.

In summary, our visualizations showcase precise localization capabilities, outperforming current SOTAs and pixel-wise reconstruction techniques. Both quantitative and qualitative results demonstrate the contribution of patch-wise constraints for distinguishing anomalies while modeling normal tissue patterns. The analysis of data from three different modalities further verifies the general applicability of the methodology.

4.4. Ablation study and analysis

4.4.1. Patch-wise loss based on contrastive learning

The proposed patch-wise loss aims to provide supervision on local image information, enabling the model to better learn the local semantic patterns of normal samples, thereby improving the anomaly detection performance of reconstruction-based approaches. To validate the effectiveness of the patch-wise loss, we conduct ablation studies qualitatively and quantitatively.

Visual evaluation. For qualitative evaluation, we visualize the reconstructions produced by models trained with and without the patch-wise loss. Specifically, the auto-encoder employs pixel-wise loss as a baseline model. As shown in Fig. 6 , the baseline model using pixel-wise and image-wise losses produced reconstructed outputs that retained lension regions (*i.e.*, tumor). However, the model trained with patch-wise and image-wise losses generated reconstructions without the lesion areas. It is worth noting that since anomaly detection is achieved by comparing the reconstructed outputs with the raw input images, we expect the method only reconstructs normal regions given the abnormal samples. Taking the first row image in Fig. 6 as an example, the reconstruction output based on pixel loss maintains consistent grayscale intensities in the image, however, the fine detail structures are lost and the lesion area is still retained. The output from pixel and image-wise losses becomes slightly clearer in details but faint lesion areas can still be observed. In contrast, the proposed reconstruction with patch-wise and image-wise losses achieves much sharper local textures. Notably, our reconstructed results predicted normal tissue textures at positions corresponding to the anomalies. This demonstrates that the patch-wise contrastive learning loss enables auto-encoders to learn the local manifold of normal samples better compared with pixel-wise losses.

Quantitative evaluation. We evaluate anomaly detection performance when incorporating patch-wise loss of the proposed approach on three datasets. As demonstrated in Table 7, adding patch-wise supervision (*i.e.*, L_{Patch}) significantly boosts model performance compared with the baseline trained with only pixel-wise and image-wise loss

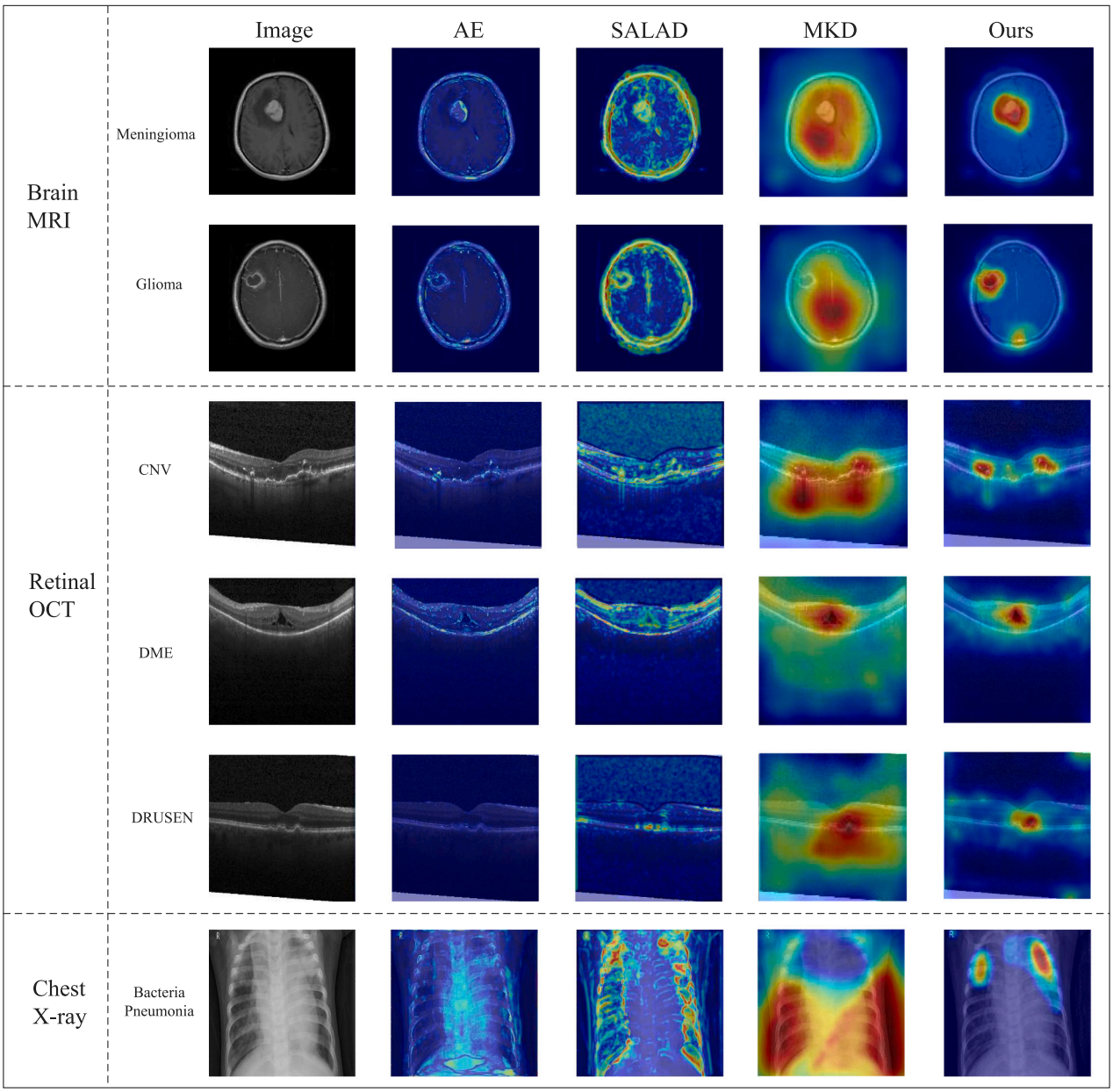

Fig. 5. Visualized heatmaps indicating potential lesion locations are presented for brain MRI, retinal OCT, and chest X-ray. In these heatmaps, red-colored regions correspond to areas identified as probable lesions. The first column shows the original medical images, while subsequent columns display lesion localization heatmaps generated by several techniques, namely auto-encoder (AE), SALAD, MKD, and our method, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

(*i.e.*, L_{Pixel} and L_{Image}). The consistent improvements validate that the patch-wise loss enables more effective modeling of normal samples by enhancing the learning of discriminative local semantic features.

In summary, the proposed patch-wise contrastive loss enables more effective learning of local semantic features for anomaly detection. Both qualitative and quantitative analyses demonstrate that contrastive learning is highly effective for supervision of patch-level representations.

4.4.2. Patch-wise anomaly score

The proposed patch-wise anomaly score is designed to contrast local semantic discrepancies between the input and output patches. This provides a robust alternative to noise-sensitive pixel-wise reconstruction errors. To validate the effectiveness of the patch-wise anomaly score, we conduct both qualitative and quantitative evaluations.

Visual evaluation. We qualitatively compared the performance of pixel-wise anomaly score and patch-wise anomaly score based on the input and output of patchCL-AE. As shown in Fig. 7 (C-D), we visualized anomaly heatmaps based on pixel-wise anomaly score and patch-wise anomaly score, respectively. It can be observed that anomaly heatmaps based on pixel-wise anomaly score are easily affected by high-frequency noise. This indicates the sensitivity of pixel-wise anomaly score to subtle inconsistencies and distortions. In contrast, heatmaps based on patch-wise anomaly score highlight the abnormal areas corresponding to the lesions clearly, while suppressing background noise. The proposed patch-wise anomaly score identifies anomalous regions based on semantic inconsistencies rather than minor pixel-level noise. This verifies that patch-wise comparisons provide more robust anomaly localization compared with pixel-wise comparisons.

Quantitative analysis. We evaluate the patch-wise and pixel-wise score methods using AUC, F1-score, ACC, SEN, and SPE. As shown in

Table 7
Ablation study on losses and anomaly scores on Brain MRI, Retinal OCT, and Chest X-ray datasets with AUC (%), F1-score (%), ACC (%), SEN (%), and SPE (%).

Datasets	Loss / Anomaly score	L_{Pixel} $+S_{Pixel}$	$L_{Pixel}+L_{Image}$	$L_{Image}+L_{Patch}$	$L_{Image}+L_{Patch}$ $+S_{patch}$
Brain MRI	AUC	76.95	91.66	96.56	**99.18**
	F1-score	87.05	91.03	93.93	**98.02**
	ACC	80.01	87.03	91.45	**97.25**
	SEN	97.24	95.25	95.69	**98.56**
	SPE	41.48	68.64	81.97	**94.32**
Retinal OCT	AUC	74.57	88.58	97.49	**99.31**
	F1-score	88.83	91.54	95.59	**99.20**
	ACC	81.50	86.70	93.80	**98.80**
	SEN	98.13	96.00	98.00	**99.86**
	SPE	31.59	58.80	81.20	**95.60**
Chest X-ray	AUC	60.02	76.19	85.06	**91.12**
	F1-score	78.25	82.81	85.91	**88.62**
	ACC	67.30	74.51	81.08	**84.77**
	SEN	94.10	**98.20**	92.30	94.87
	SPE	22.64	35.04	62.39	**67.94**

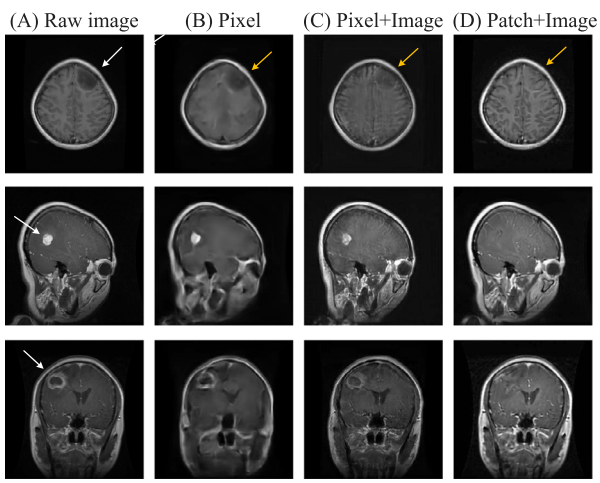

Fig. 6. Reconstructed outputs from the auto-encoder using different loss functions. (A) Raw image: the first column shows three different brain images. (B) Pixel loss: the second column shows reconstructions using the auto-encoder with pixel-wise loss. (C) Pixel+Image loss: the third column shows reconstructions using pixel-wise and image-wise losses. (D) Patch+Image loss: the last column shows reconstructions using the proposed patch-wise and image-wise loss.

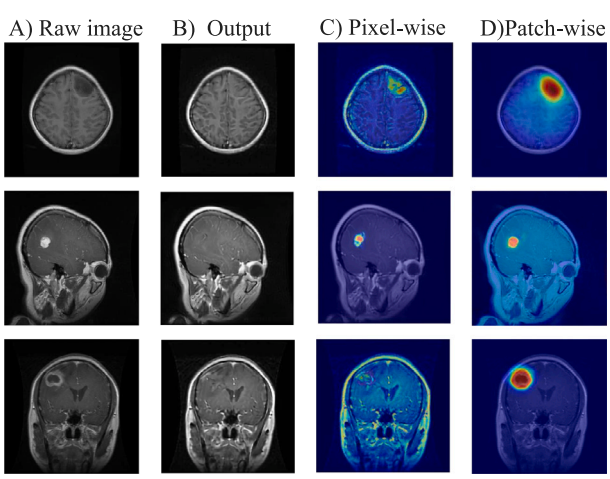

Fig. 7. Visualization heatmaps from the auto-encoder using different anomaly scores. (A) Raw image: the first column shows three different brain images. (B) Reconstruction output: the second column shows the reconstruction outputs of patchCL-AE. (C) Pixel-wise anomaly score: the third column shows heatmaps using pixel-wise anomaly score. (D) Patch-wise anomaly score: the fourth column shows heatmaps using patch-wise anomaly score.

Table 7, our method using patch-wise score (S_{patch}) achieves superior performance across all metrics compared with the pixel-wise score (S_{pixel}). Notably, using S_{patch} surpasses an AUC of 99% on brain and retinal images, demonstrating reliable anomaly discrimination capabilities. This highlights the benefits of leveraging patch-wise semantic comparisons beyond just pixel-wise reconstruction errors.

In total, these qualitative and quantitative results validate the advantages of the proposed patch-wise anomaly score. By focusing on local semantic deviations rather than pixel statistics, patch-wise score enables more discriminative and robust anomaly detection.

4.4.3. *Effectiveness of contrastive learning for patch-wise loss*

The improvements in patch-wise loss stem from contrastive learning's ability to optimize local discrimination during training. By explicitly maximizing the consistency between patch representations of the input and output, contrastive learning in the loss function provides the necessary localized supervision at the patch level. This induces more fine-grained distinction of normal local patterns, which is critical for precise anomaly detection. To empirically validate the above statement, we conduct an ablation study comparing models trained with and without contrastive learning loss.

There are patch-level losses based on mean squared error (MSE) and cosine similarity, which primarily aims to bring corresponding patches in the input and output closer together. In contrast, the proposed patch-wise contrastive loss contains two components: (1) pulling corresponding patches in the input and output closer, and (2) pushing non-corresponding patches farther apart. We train three auto-encoder models on medical images, where the first one uses a MSE-based patch loss, the second one uses cosine-based patch loss, and the last one uses the proposed patch-wise contrastive learning (CL) loss.

As shown in Table 8, the model using patch-wise CL loss achieves higher performance across all three datasets compared with using MSE or cosine-based patch loss. These results validate that maximizing the consistency between positive patch pairs while minimizing the consistency between negative pairs helps learn more robust and effective feature representations, hence improving anomaly detection performance. Rather than simply reducing the distance between corresponding patches, contrastive learning loss optimizes patch embeddings better by distinguishing between matching and non-matching pairs.

5. Conclusion

A novel patch-wise contrastive learning-based auto-encoder is proposed for more accurate anomaly detection. The proposed patch-wise contrastive loss provides local semantic supervision on the auto-encoder's reconstruction, overcoming the lack of localization capability in previous methods. In addition, we introduce a local semantic

Table 8
Ablation study on different patch-wise losses on Brain MRI, Retinal OCT, and Chest X-ray datasets with AUC (%), F1-score (%), ACC (%), SEN (%), and SPE (%).

Datasets	Metrics	patchCL-AE		
		cosine	MSE	Our CL
Brain MRI	AUC	94.67	92.93	**99.18**
	F1-score	97.30	96.03	**98.02**
	ACC	96.18	94.35	**97.25**
	SEN	**99.66**	99.00	98.56
	SPE	88.39	83.95	**94.32**
Retinal OCT	AUC	90.98	91.34	**99.31**
	F1-score	91.97	91.27	**99.20**
	ACC	87.50	86.30	**98.80**
	SEN	95.46	95.60	**99.86**
	SPE	63.60	58.40	**95.60**
Chest X-ray	AUC	85.73	84.63	**91.12**
	F1-score	86.12	86.13	**88.62**
	ACC	81.25	80.28	**84.77**
	SEN	93.07	**97.94**	94.87
	SPE	61.53	50.85	**67.94**

discrepancy-based anomaly score that compares local semantic difference instead of pixel-level errors. Experiments validate that our patch-wise contrastive loss guides the learning of localized semantics better, enabling more accurate anomaly detection. The proposed patch-wise anomaly score can locate lesions more precisely without being affected by pixel-level noise. We have verified the robustness of our approach on three public datasets and achieved state-of-the-art results. Overall, this research demonstrates the efficiency of leveraging patch-wise contrastive learning for anomaly detection and diagnosis.

Our method assumes large semantic discrepancies across patches, suitable for anomaly detection in medical images with complex textures, but may not generalize well for images with repetitive local patterns. In future work, we plan to relax this assumption to accommodate more diverse medical image datasets. In addition, an important potential for anomaly detection is the identification of rare diseases, which is very important in clinical research. Taking the known diseases as a group of normal (known diseases) images, the anomaly detection algorithm identifies the images different from the normal (known diseases) to find new diseases or rare diseases. Our future work will extend to this topic.

CRediT authorship contribution statement

Shuai Lu: Writing – original draft. **Weihang Zhang:** Writing – review & editing. **Jia Guo:** Methodology. **Hanruo Liu:** Resources. **Huiqi Li:** Project administration, Writing – review & editing. **Ningli Wang:** Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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